

ROY CARMELLI

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SUMMARY

Second-year B.Sc. student in Computer Science & Neuroscience at Bar-Ilan University - a program combining software engineering depth with hands-on neuroscience research. Independently ships web apps, Chrome extensions, and AI-powered tools. utilizing Agentic workflow development as a core methodology. Seeking student/intern role, with high ambition to learn and grow.

EDUCATION

B.Sc. Computer Science & Neuroscience | Bar-Ilan University | GPA: 86.16

Oct 2024 – present

Relevant coursework: Data Structures & Algorithms (93) · OOP in Java (92) · Computer Architecture (C/Assembly) · Neuroscience (94) · Biochemistry (99) · Calculus (100) · Linear Algebra (97) · Python Data Processing (98)

PROJECTS

CareerPredict AI | React 19 · Node.js · Express · SQLite · Google OAuth · Render [↗ live demo](#) [↗ GitHub](#)

- A web app that generates a personalized AI career portrait based on user onboarding. Full-stack SPA - Google OAuth 2.0 login → multi-step onboarding → AI-generated career portrait via Pollinations.ai (100+ keyword-classified outcomes). React/Vite/Tailwind frontend, Node/Express + Passport.js backend, SQLite - production-deployed on Render with full end-to-end ownership from DB schema to deployment

Wine Sommelier Telegram Bot | Python · Google Gemini API · Telegram API · Google Sheets · Vercel [↗ GitHub](#)

- A Telegram bot acting as a personal sommelier - recommends wine pairings from a live cellar inventory, advises on purchases, explains wine education, and holds open-ended sommelier conversations. Serverless webhook from Google Sheets; built resilient 4-model fallback chain with exponential backoff after identifying reliability gaps in production. Two-layer memory: active session history (30 msgs, 1-hr TTL) + AI-compressed long-term summary

AI Sidebar - Chrome Extension | JavaScript · Chrome Extensions API · Manifest V3 [↗ GitHub](#)

- A Chrome extension that injects an AI chat sidebar into any webpage, Integrates 6 AI API providers via Factory + Strategy + Template Method OOP patterns; Service Worker (MV3) with cross-origin postMessage security and custom Markdown-to-HTML renderer

Personal Portfolio | React 18 · TypeScript · Vite · Tailwind CSS · Framer Motion [↗ live site](#) [↗ GitHub](#)

- Portfolio site with an embedded endless runner game engine - TypeScript OOP, AABB collision, physics simulation, 7-state FSM, and Custom Event Bus for zero-prop-drilling component communication

CLR - Culinary Logic Repository | TypeScript · Python · Groq API · Telegram API · Vercel [↗ GitHub \(in active development\)](#)

- A culinary content system, enhancing my culinary passion. AI ingestion pipeline: URL input via Telegram → Groq LLM extracts and structures content → normalized JSON consumed by React web app - engineered utilizing automated agentic workflows for rapid prototyping and deployment

MILITARY SERVICE

IDF Combat Medic & Medical Coordinator | Gaza Division

Nov 2021 – present

- Operational Leadership:** Directed a specialized medical unit, managing a team that scaled from 12 to 30+ personnel during high-intensity conflict; oversaw complex logistics and personnel deployment under extreme pressure.
- Operations Coordination:** Managed real-time data synchronization and communication protocols within a divisional war room to ensure rapid response and resource optimization across multiple units.
- Crisis Resilience & Decision Making:** Orchestrated high-stakes triage and evacuation logistics during mass-casualty events, demonstrating extreme resilience and technical decision-making clarity in active-threat environments.

SKILLS

Languages: Java 17, Python, C, Assembly

Data & Analysis: NumPy, Pandas, Matplotlib, Jupyter

Tools & Platforms: Git/GitHub, Chrome Extensions API (MV3), Render, Vercel, GitHub Pages

AI-Assisted Dev & MCP: Architecture, implementation, debugging, and code review.

Certifications: [↗ Claude Code in Action](#) | [↗ Claude Code 101](#) | [↗ Claude 101](#)

LEADERSHIP & ADDITIONAL

StandWithUs Fellowship (2025) - Selected for competitive Israel advocacy and leadership fellowship; trained in public speaking, media engagement, and cross-cultural communication

Languages: Hebrew (native), English (professional proficiency)